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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECT REPORT ON

SUPERMARKET CHECKOUT AND DISCOUNTS BILLING SYSTEM

Submitted by

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APRIL 2026

Semester: II

DECLARATION

I hereby declare that the project entitled "**SUPERMARKET CHECKOUT AND DISCOUNTS BILLING SYSTEM**" submitted in partial fulfillment of the requirements for the course **Java Programming language** is a **record of my original work** carried out under the guidance of **DR.BOOPALAN SUDARAMOORTHY**, Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

HARSHINI.M

CERTIFICATE

This is to certify that the project entitled **"SUPERMARKET CHECKOUT AND DISCOUNTS BILLING SYSTEM "** is a **bona fide work** carried out by **Harshini.M,BSC CS 2025-2026**, BSC.Computer science, **KG College of Arts & Science**, in partial fulfillment of the requirements for the course **Java Programming language**.

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place: Coimbatore

Date:10/04/2026

Submitted for the Viva-Voce Examination held on

_____10/04/2026_____

Guide:

(Technical Trainer, KGCAS)

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Head of the Department

(Technical Head, KGCAS)

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TABLE OF CONTENTS

S. No.	Topics	Page. No.
1	**Introduction**	
1.1	Objective	
1.2	Overview	
1.3	Features	
2	**Program Analysis**	
2.1	Problem Definition	
2.2	Project Overview	
2.3	Scopes and Limitations	
3	**System Design**	
3.1	Modules	

4	**Methodology**	
4.1	Problem Analysis	
4.2	System Design	
4.3	Program Development	
4.4	Testing and Validation	
4.5	Result and Documentation	
4.6	Algorithm	
4.7	Flowchart	
5	**Source Code**	
6	**Input and Output Screens**	
7	**Discussion and Conclusion**	
7.1	Discussion	

7.2	Conclusion	
8	**References**	

ABSTRACT

The **Supermarket Checkout And Discount Billing System** is a Java-based application developed using **Swing GUI** to simplify and automate the billing process in a supermarket. The system allows users to enter product details such as name, price, and discount percentage, and generates a bill in a structured format using a table layout.

The application is designed using core Java concepts such as **classes and objects** to represent products and billing, **encapsulation** for secure data handling, **inheritance and polymorphism** to manage discounted products, and **exception handling** to handle invalid inputs. The system also uses **file handling** to store billing records for future reference.

The graphical user interface provides a user-friendly environment where products can be added dynamically, and the total bill is calculated automatically. The use of a table structure enhances readability and mimics real-world supermarket billing systems.

This project demonstrates the practical implementation of Java programming concepts in developing a real-time application, making it useful for small retail stores to manage billing efficiently and accurately

1. INTRODUCTION

In modern retail environments, manual billing systems are inefficient and prone to errors. To improve accuracy and speed, computerized billing systems are widely used in supermarkets. The **Supermarket Checkout And Discount Billing System** is developed to automate the billing process and make it faster, more reliable, and user-friendly.

This project is implemented using Java programming language and **Swing GUI**, which provides an interactive interface for users to enter product details such as product name, price, and discount percentage. The system calculates the final price after applying the discount and generates a complete bill in a structured format using a table.

The application uses core Java concepts such as classes and objects for designing the system, encapsulation for data security, inheritance and polymorphism for handling product variations, and exception handling to manage invalid inputs. File handling is also used to store billing details for future reference.

Overall, this system reduces manual effort, minimizes calculation errors, and provides an efficient solution for supermarket billing operations.

1.1 Objective

The main objective of the **Supermarket Checkout And Discount Billing System** is to develop a simple and efficient billing application using Core Java that automates the process of discount and bill generation in supermarkets

The specific objectives of this project are:

- To develop a user-friendly billing system using Java and Swing GUI.
- To automate the process of calculating product prices with discounts.
- To reduce manual errors in billing and improve accuracy.
- To implement core Java concepts such as classes, objects, inheritance, polymorphism, and encapsulation.
- To handle invalid user inputs using exception handling techniques.
- To store billing details using file handling for future reference.
- To display the bill in a structured table format similar to real supermarket systems.
- To improve efficiency and speed in the billing process

1.2 Overview

The **Supermarket Checkout And Discount Billing System** is a Java-based desktop application designed to simplify and automate the billing process in supermarkets. It provides a graphical user interface using Java Swing, allowing users to easily enter product details such as product name, price, and discount percentage.

The system processes the entered data and calculates the final price after applying the discount. All products are displayed in a table format, which resembles a real supermarket billing system. Users can add multiple items, and the system automatically calculates the total amount for all products.

The application is developed using core Java concepts such as object-oriented programming, including classes, objects, inheritance, polymorphism, and encapsulation. It also uses exception handling to manage invalid inputs and file handling to store billing records.

Overall, the system provides a simple, efficient, and user-friendly solution for handling supermarket billing operations, making it suitable for small retail environments.

1.3 Features

Features – Supermarket Checkout and Discount Billing System

1. **User-Friendly Interface**
   Provides an easy-to-use graphical interface using Java Swing for smooth interaction.
2. **Product Entry System**
   Allows users to enter product name, price, and discount percentage during checkout.
3. **Table-Based Billing**
   Displays all added items in a structured table format like a real supermarket bill.
4. **Automatic Discount Calculation**
   Calculates the final price of each product after applying the discount automatically.
5. **Multiple Item Billing**
   Supports adding multiple products and generates a combined total bill.
6. **Automatic Total Calculation**
   Computes the total amount of all items instantly.
7. **Error Handling**
   Handles invalid inputs (like letters instead of numbers) using exception handling.
8. **File Storage**
   Saves billing details into a file (bill.txt) for future reference.
9. **Fast and Efficient Processing**
   Reduces manual work and speeds up the checkout process.
10. **Real-Time Output Display**
    Shows billing details instantly on the screen as items are added.

1. PROGRAM ANALYSIS

Program analysis is an important phase in software development where the problem is studied in detail and the requirements of the system are identified. This stage helps in understanding how the system should work and what features are required to solve the problem efficiently.

The analysis phase focuses on identifying the current issues in the existing system and proposing a computerized solution that improves efficiency and accuracy.

2.1 Problem Definition

In traditional supermarket billing systems, manual calculations are time-consuming and prone to errors. Managing multiple products, applying discounts, and calculating the total bill manually can lead to inaccuracies and inefficiency. There is a need for an automated system that can handle product entry, apply discounts correctly, and generate accurate bills quickly. This project aims to solve these problems by developing a computerized billing system.

2.2 Project Overview

The Supermarket Checkout and Discount Billing System is a Java-based application developed using Swing GUI. It allows users to enter product details such as name, price, and discount percentage during checkout. The system calculates the final price after applying discounts and displays all items in a table format similar to real supermarket billing systems. It also calculates the total bill automatically and stores the billing details using file handling. The system is designed using object-oriented programming concepts for better structure and efficiency.

This project is implemented using **Core Java programming concepts** and can be developed as either a **console-based program or a Java Swing GUI application**.

2.3 Scopes and Limitations

Scope

The scope of the **Supermarket Checkout And Discount Billing System** includes the following:

- Can be used in small supermarkets and retail shops
- Supports multiple product entries
- Provides automatic discount calculation
- Generates a structured bill in table format
- Stores billing records for future reference
- Easy to use with a graphical user interface

Limitations

- Limited to basic billing functions (no barcode scanning)
- Does not support database storage (uses only file storage)

- No multi-user or login system
- Suitable only for small-scale applications
- No online payment integration

2.SYSTEM DESIGN

System design is the stage where the structure and components of the system are planned before the actual implementation. In this phase, the system is divided into different modules, and the workflow of the application is defined. Proper system design helps in organizing the program efficiently and makes development easier.

The **Supermarket Checkout and Discount Billing System** is designed as a GUI-based application where users can enter product details, apply discounts, and generate a final bill automatically.

The system is divided into several modules to perform specific tasks. Each module is responsible for handling a particular function of the system.

🔹 3.1 Modules

The main modules of the Supermarket Checkout and Discount Billing System are explained below:

1. Product Entry Module

This module is responsible for entering product details such as product name, price, and discount percentage.

Example Input:

Product Name: Rice
Price: ₹100
Discount: 10%

The user enters these details through the graphical interface.

2. Item Addition Module

This module allows the user to add products to the billing table.

Functions of this module include:

- Accepting product details
- Calculating final price after discount
- Adding product details to the table
- Allowing multiple items to be added

The system supports adding multiple products before generating the final bill.

3. Bill Calculation Module

This module performs all the billing calculations.

The system calculates:

- Final price of each product after discount
- Total amount of all products

The calculation is performed automatically using Java logic.

Example Calculation:

Final Price = Price – (Price × Discount / 100)
Total Bill = Sum of all final prices

4. Bill Display Module

This module displays the bill in a structured table format.

The bill includes:

- Product Name
- Price
- Discount (%)
- Final Price

It also displays the total bill amount at the bottom.

Example Output:

```
------ SUPERMARKET BILL ------
Item      Price   Discount   Final Price
Rice      100     10%        90
Milk      50      5%         47.5

Total: 137.5
```

5. File Storage Module

This module saves the billing details into a file (bill.txt).

It helps in:

- Storing bill records
- Viewing previous transactions

6. Exit Module

This module allows the user to exit the application safely after completing the billing process.

When the user closes the window, the system terminates the program.

3.METHODOLOGY

Methodology describes the process followed to develop the project. It explains the steps taken from understanding the problem to designing, implementing, testing, and documenting the system.

For the Supermarket Checkout and Discount Billing System, a structured development approach was followed to ensure that the system meets the requirements and functions efficiently.

The development process includes the following stages:

• Problem Analysis
• System Design
• Program Development
• Testing and Validation
• Result and Documentation

4.1. Problem Analysis

In this phase, the problem is studied and analyzed in detail. The requirements of the system are identified based on the needs of supermarket billing operations.

The main problems identified in manual billing systems are:

- Manual calculations may lead to billing errors
- Preparing bills manually takes more time
- Difficult to handle multiple products efficiently
- No proper record of billing transactions

Based on these problems, the system requirements were defined to develop an automated billing application that simplifies supermarket checkout operations.

4.2. System Design

In this stage, the overall structure of the system is designed before coding begins.

The system is divided into modules such as:

- Product Entry Module
- Item Addition Module
- Bill Calculation Module

- Bill Display Module
- File Storage Module
- Exit Module

The program flow and structure are designed so that users can easily interact with the system through a graphical interface.

4.3. Program Development

After designing the system, the program is implemented using the Java programming language with Swing GUI.

The following steps were followed during development:

- Creation of classes and methods
- Implementation of GUI interface
- Writing code for discount calculation
- Handling user input
- Displaying product details in table format
- Generating total bill

Core Java concepts such as classes, objects, loops, conditional statements, methods, inheritance, and exception handling were used during developme

.

4.4. Testing and Validation

Testing ensures that the program works correctly and produces the expected results.

Different test cases were used to check the functionality of the system.

Example Test Cases:

Test Case	Expected Result
Enter product details	Product added successfully

Add multiple items	All items displayed in table
Generate total	Correct total amount displayed
Invalid input	Error handled properly

This phase ensures that the program runs without errors and produces accurate results.

4.5. Result and Documentation

After successful testing, the results of the system are documented. The system generates the final bill based on the entered product details and discounts.

The results are demonstrated through program execution and screenshots showing input and output.

Proper documentation is prepared to explain the design, implementation, and working of the system.

4.6. Algorithm

Algorithm for the Supermarket Checkout and Discount Billing System:

1. Start the program
2. Enter product name, price, and discount
3. Calculate final price using discount formula
4. Add product details to the table
5. Repeat steps 2–4 for multiple items
6. Calculate total bill amount
7. Display the bill in table format
8. Save the bill to file
9. End the program

4.7. Flowchart

Students should draw a flowchart showing program flow.

Example Flow:

Start

↓

Enter Product Details

↓

Add to Table

↓

Calculate Total

↓

Display Bill

↓

End

Students can draw using:

- draw.io
- Lucidchart
- MS Visio

1. SOURCE CODE

Include important parts of the Java code such as:

Main Class

```
public class SupermarketGUI {
    public static void main(String[] args) {
        new SupermarketGUI();
    }
}
```

Product Entry Logic

```
String name = nameField.getText();
double price = Double.parseDouble(priceField.getText());
double discount = Double.parseDouble(discountField.getText());
```

Billing Calculation Logic

```
double finalPrice = price - (price * discount / 100);
```

Total Calculation

```
double total = 0;
for (int i = 0; i < model.getRowCount(); i++) {
    total += (double) model.getValueAt(i, 3);
}
```

File Handling

```
FileWriter fw = new FileWriter("bill.txt");
fw.write("Total: " + total);
fw.close();
```

INPUT AND OUTPUT SCREENS

Figure 1: Main Screen

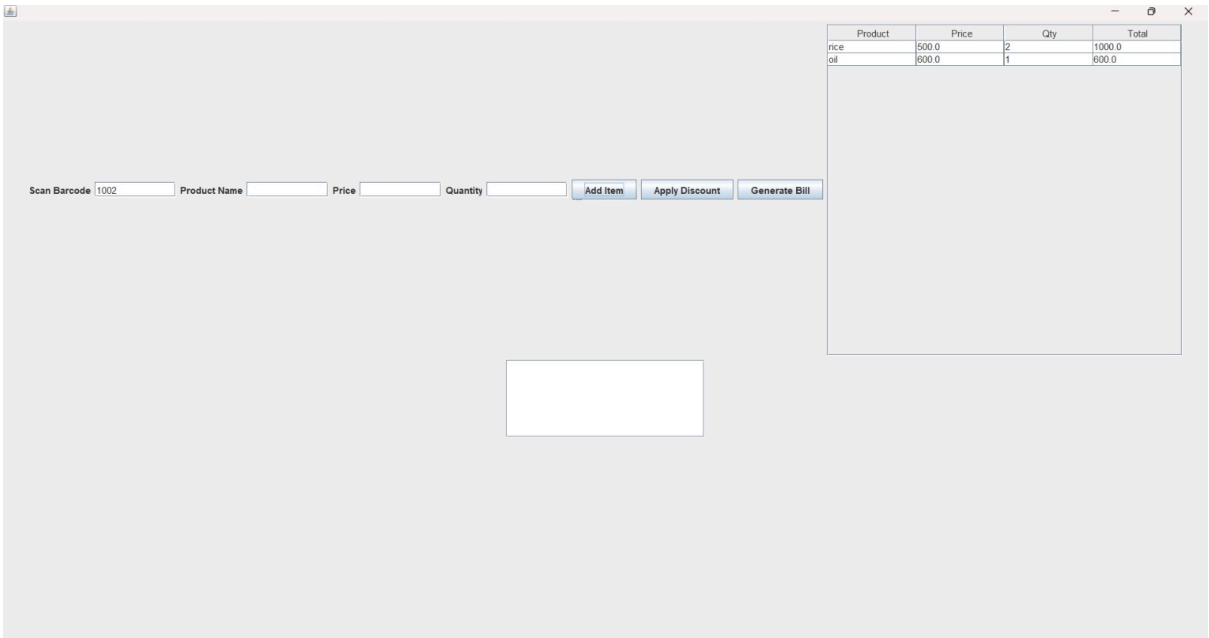


Figure 2:Adding Discount

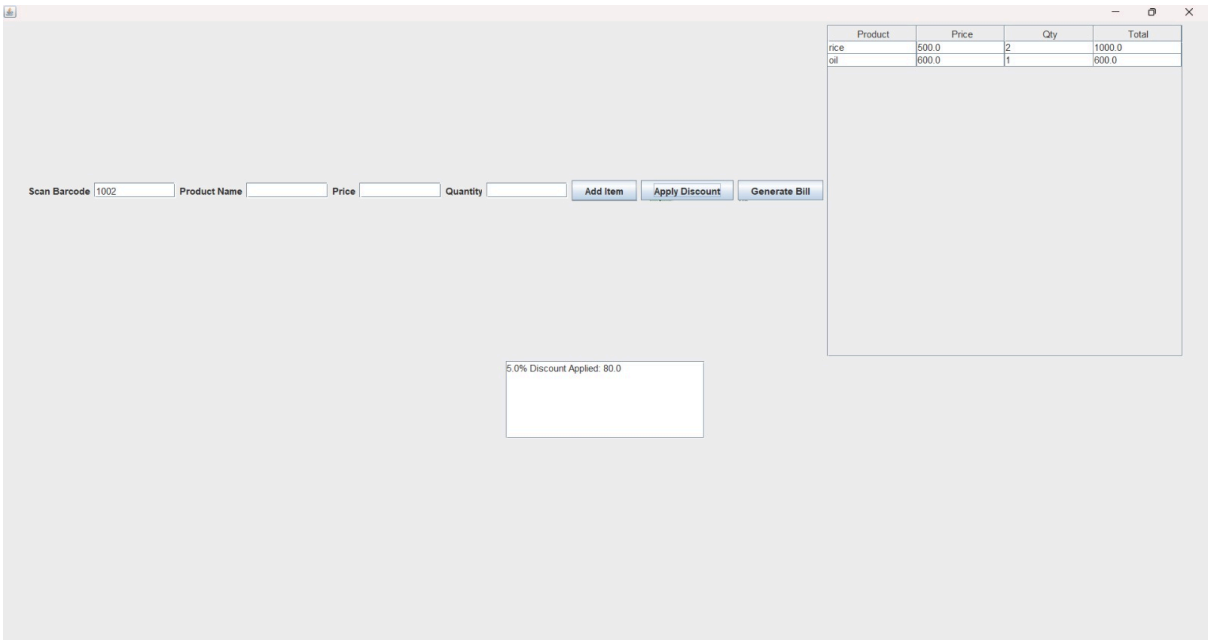
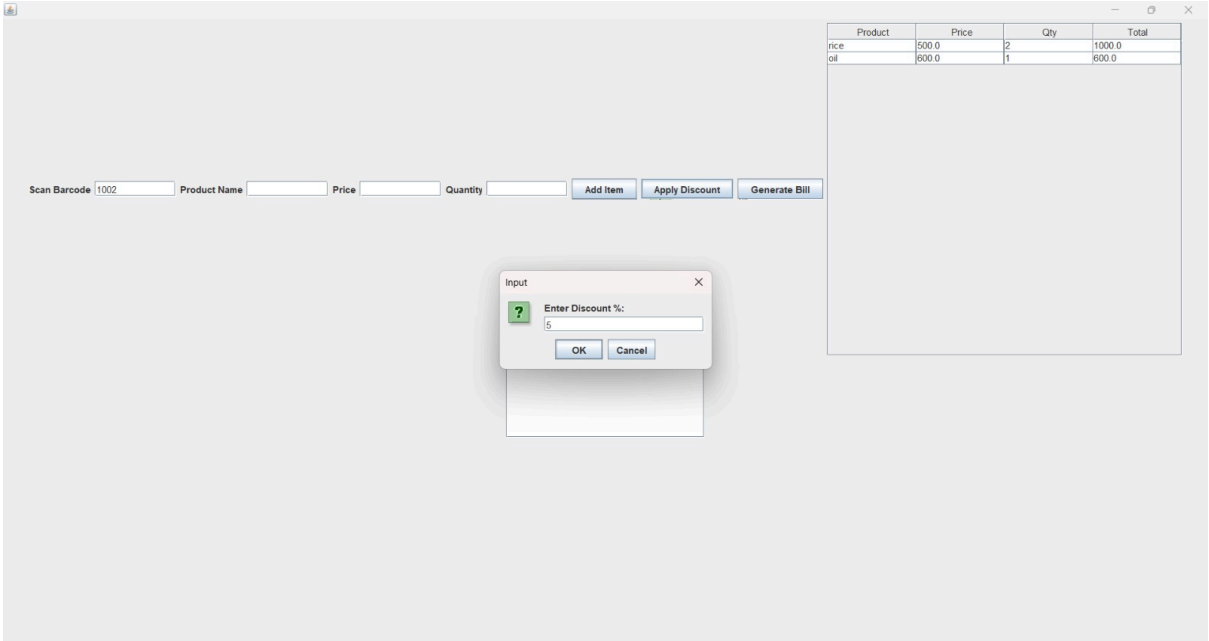


Figure 3: Discount added



DISCUSSION & CONCLUSION

7.1 Discussion

The **Supermarket Checkout and Discount Billing System** was developed to automate the checkout and billing process in supermarkets. The system allows users to enter product details such as name, price, and discount, and generates the bill automatically in a structured table format.

During the development process, Core Java programming concepts such as classes, objects, inheritance, polymorphism, loops, conditional statements, and exception handling were applied. The system successfully performs discount calculations and generates accurate billing details.

The system was tested with different inputs to ensure correct functionality. The results show that the system reduces manual effort, minimizes errors, and improves billing efficiency.

7.2 Conclusion

The **Supermarket Checkout and Discount Billing System** successfully demonstrates the practical application of Core Java programming concepts in developing a real-world application.

The system automates the billing process, applies discounts accurately, and generates bills quickly. It improves efficiency and reduces manual errors in supermarket checkout operations.

This project helps in understanding how Java can be used to develop user-friendly and efficient software systems. The system can be further enhanced by adding features such as database integration, barcode scanning, and online payment options.

REFERENCES

The following resources were used during the development of this project:

Books

- Herbert Schildt – *Java: The Complete Reference*

Websites

- https://www.oracle.com/java
- https://www.geeksforgeeks.org
- https://www.javatpoint.com

1. INPUT AND OUTPUT SCREENS

Include screenshots of:

- Program menu
- Item selection
- Billing output
- Final invoice

Each screenshot should be labeled.

Example:

Figure 1: Menu ScreenFigure 2: Bill Output

1. DISCUSSION & CONCLUSION

7.1 Discussion

The **Hotel & Restaurant Billing System** was developed to automate the billing process in restaurants and hotels. The system allows users to select food items, enter quantities, and generate bills automatically.

During the development process, Core Java programming concepts such as loops, conditional statements, methods, and object-oriented principles were applied. The system successfully performs bill calculations and generates accurate invoices.

The system was tested with different inputs to ensure that it performs the required calculations correctly. The results show that the system reduces manual effort and improves billing efficiency.

7.2 Conclusion

The **Hotel & Restaurant Billing System** successfully demonstrates the practical application of Core Java programming concepts in developing a real-world business application.

The system automates the billing process, reduces calculation errors, and provides faster service to customers. It simplifies restaurant billing operations by generating accurate invoices based on customer orders.

This project helps students understand how Java programming can be used to develop practical software systems. The project can be further enhanced by adding features such as database connectivity, graphical user interfaces, and online payment systems.

1. REFERENCES

The following resources were used during the development of this project:

Books

• Herbert Schildt – *Java: The Complete Reference*

Websites

• https://www.oracle.com/java• https://www.geeksforgeeks.org• https://www.w3schools.com• https://www.javatpoint.com